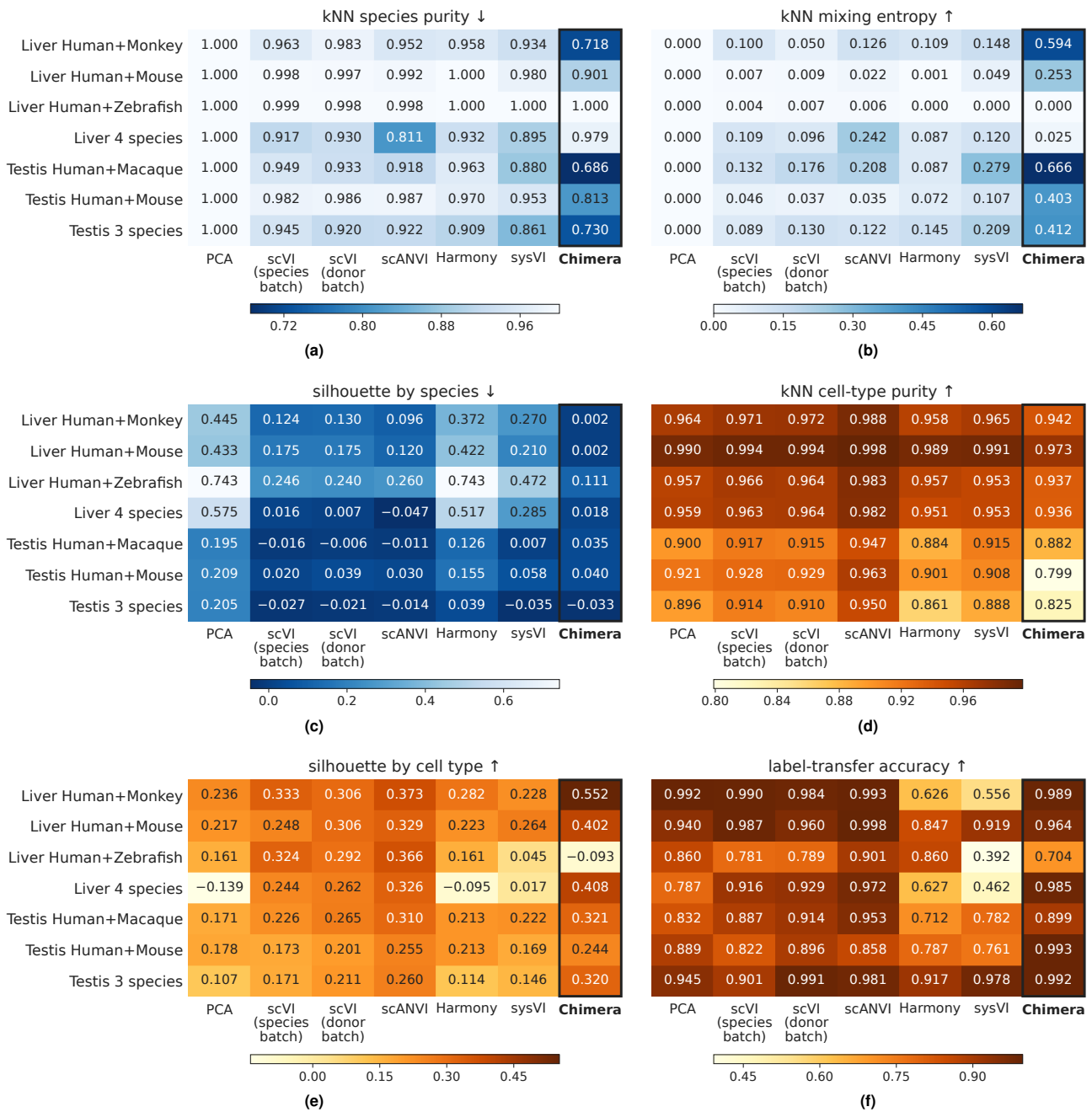

kNN species purity ↓
Liver Human+Monkey 1.000 0.963 0.983 0.952 0.958 0.934 0.718
Liver Human+Mouse 1.000 0.998 0.997 0.992 1.000 0.980 0.901
Liver Human+Zebrafish 1.000 0.999 0.998 0.998 1.000 1.000 1.000
Liver 4 species 1.000 0.917 0.930 0.811 0.932 0.895 0.979
Testis Human+Macaque 1.000 0.949 0.933 0.918 0.963 0.880 0.686
Testis Human+Mouse 1.000 0.982 0.986 0.987 0.970 0.953 0.813
Testis 3 species 1.000 0.945 0.920 0.922 0.909 0.861 0.730
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.72 0.80 0.88 0.96
(a)
kNN mixing entropy ↑
0.000 0.100 0.050 0.126 0.109 0.148 0.594
0.000 0.007 0.009 0.022 0.001 0.049 0.253
0.000 0.004 0.007 0.006 0.000 0.000 0.000
0.000 0.109 0.096 0.242 0.087 0.120 0.025
0.000 0.132 0.176 0.208 0.087 0.279 0.666
0.000 0.046 0.037 0.035 0.072 0.107 0.403
0.000 0.089 0.130 0.122 0.145 0.209 0.412
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45 0.60
(b)
silhouette by species ↓
Liver Human+Monkey 0.445 0.124 0.130 0.096 0.372 0.270 0.002
Liver Human+Mouse 0.433 0.175 0.175 0.120 0.422 0.210 0.002
Liver Human+Zebrafish 0.743 0.246 0.240 0.260 0.743 0.472 0.111
Liver 4 species 0.575 0.016 0.007 −0.047 0.517 0.285 0.018
Testis Human+Macaque 0.195 −0.016 −0.006 −0.011 0.126 0.007 0.035
Testis Human+Mouse 0.209 0.020 0.039 0.030 0.155 0.058 0.040
Testis 3 species 0.205 −0.027 −0.021 −0.014 0.039 −0.035 −0.033
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.0 0.2 0.4 0.6
(c)
kNN cell-type purity ↑
0.964 0.971 0.972 0.988 0.958 0.965 0.942
0.990 0.994 0.994 0.998 0.989 0.991 0.973
0.957 0.966 0.964 0.983 0.957 0.953 0.937
0.959 0.963 0.964 0.982 0.951 0.953 0.936
0.900 0.917 0.915 0.947 0.884 0.915 0.882
0.921 0.928 0.929 0.963 0.901 0.908 0.799
0.896 0.914 0.910 0.950 0.861 0.888 0.825
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.80 0.84 0.88 0.92 0.96
(d)
silhouette by cell type ↑
Liver Human+Monkey 0.236 0.333 0.306 0.373 0.282 0.228 0.552
Liver Human+Mouse 0.217 0.248 0.306 0.329 0.223 0.264 0.402
Liver Human+Zebrafish 0.161 0.324 0.292 0.366 0.161 0.045 −0.093
Liver 4 species −0.139 0.244 0.262 0.326 −0.095 0.017 0.408
Testis Human+Macaque 0.171 0.226 0.265 0.310 0.213 0.222 0.321
Testis Human+Mouse 0.178 0.173 0.201 0.255 0.213 0.169 0.244
Testis 3 species 0.107 0.171 0.211 0.260 0.114 0.146 0.320
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45
(e)
label-transfer accuracy ↑
0.992 0.990 0.984 0.993 0.626 0.556 0.989
0.940 0.987 0.960 0.998 0.847 0.919 0.964
0.860 0.781 0.789 0.901 0.860 0.392 0.704
0.787 0.916 0.929 0.972 0.627 0.462 0.985
0.832 0.887 0.914 0.953 0.712 0.782 0.899
0.889 0.822 0.896 0.858 0.787 0.761 0.993
0.945 0.901 0.991 0.981 0.917 0.978 0.992
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.45 0.60 0.75 0.90
(f)